

TEZPUR UNIVERSITY
Test V, Autumn 2016
C0 504 Natural Language Processing

Marks : 25

Time : 30 minutes

Fill in the blanks using options provided or tick the suitable option. Each question or part thereof carries one mark.

1. A TreeBank is a collection of _____ (parsed sentences / corpora).
2. In a probabilistic context free grammar, the probabilities are associated with _____ (production rules / non-terminals).
3. In a probabilistic context free grammar, the probabilities are obtained from a _____ (TreeBank / lexicon).
4. In Dependency parsing relationships are identified between _____ (phrases / words).
5. Dependency parsing is suitable for free word order languages. True or False?
6. Garden path sentences are _____ (ambiguous / temporarily ambiguous).
7. The value of a *function* in First Order Predicate Calculus (FOPC) must be Boolean. True or False?
8. In representation of meaning of a sentence vagueness that exists in the sentence should be removed. _____ (True / False).
9. If $P \Rightarrow Q$ and P is False, then Q is False too. True or False?
10. Consider the sentence-

A shop near TU-Gate sells fruits.

The FOPC representation for the above sentence would be-

- (a) $\exists s \text{ ISA}(\text{shop}, s) \wedge \text{near}(\text{locationof}(s), \text{locationof}(\text{TU-Gate})) \Rightarrow \text{sells}(s, \text{fruits})$
 - (b) $\forall s \text{ ISA}(\text{shop}, s) \wedge \text{near}(\text{locationof}(s), \text{locationof}(\text{TU-Gate})) \wedge \text{sells}(s, \text{fruits})$
 - (c) $\exists s \text{ ISA}(\text{shop}, s) \wedge \text{near}(\text{locationof}(s), \text{locationof}(\text{TU-Gate})) \wedge \text{sells}(s, \text{fruits})$
 - (d) $\forall s \text{ ISA}(\text{shop}, s) \wedge \text{near}(\text{locationof}(s), \text{locationof}(\text{TU-Gate})) \Rightarrow \text{sells}(s, \text{fruits})$
11. The better FOPC representation of “Anil is a TU-student” is
(a) $\text{TU-student}(\text{Anil})$ (b) $\text{ISA}(\text{TU-student}, \text{Anil})$
 12. The FOPC expression-
 $\text{ISA}(\text{eating}, x) \wedge \text{durationof}(x, v) \wedge \text{precedes}(\text{endof}(v), \text{Now})$
where $\text{precedes}(x, y)$ means x precedes y , means
(a) Ate (b) eating (c) will eat
 13. The expression – $\lambda a \text{ eating}(\text{Anil}, a)(\text{apple})$ evaluates to $\text{eating}(\text{Anil}, \text{apple})$
True or False?

14. The statement, “A believes B is strong” can be represented in FOPC as- Believe(A, strong(B)). True or False?
15. Identify cases of homophones, homographs, homonyms, polysemy, synonyms and hyponymy in the following pairs (meanings in some cases are indicated within brackets)-
- (a) Serve (food) and serve (organisation) _____
 - (b) bass (sound) and bass (fish) _____
 - (c) table (furniture) and table (formatted data) _____
 - (d) table and furniture _____
 - (e) bank (financial) and bank (riverside) _____
 - (f) children and kids _____
 - (g) heat (verb) and hit _____
 - (h) bank (of blood) and bank (of book) _____
16. A synset in a WordNet is -
- (a) Set of synchronising tokens
 - (b) lexical database
 - (c) set of homonyms
 - (d) set of synonyms
17. Indicate in which of the following cases selection restrictions can help in word sense disambiguation-
- (a) You should not kill the spirit. (Which sense of spirit?)
 - (b) KF serves momos. (Which sense of serve?)
18. If Q and D are the binary vectors denoting the occurrence of content terms in a query and in a document, the relevance of the document to the query can be computed as-
- (a) Sum of the vectors Q and D
 - (b) Product of the vectors Q and D
 - (c) Dot product of the vectors Q and D
 - (d) Cross product of the vectors Q and D